

DATA STRUCTURES			
Course Code: CS-201			
Credit Hours: 4(3+1)			
Pre-requisite: CS-112			
Course Objectives			
The key objectives of this course include: - To understand the design of fundamental data structures as well as algorithms that operate on them. - To understand the fundamental tradeoffs in the design of the data structures. - To introduce tools for analyzing the time and space complexity of data structures.			
Course Learning Outcomes			
After successful completion of this course, the students should be able to:	PLO	BT Level	
1. Understand the fundamentals of data structure as well as algorithms that operate on them	2	C2	
2. Apply and design various data structures methods for specified applications.	3, 4	C3	
3. Analyze empirically the behavior of algorithms in terms of time and memory used	3	C4	
Course Contents			
Abstract data types, complexity analysis, Big Oh notation, Arrays and its operations, linked list & its various types, Stacks (linked lists and array implementations), divide and conquer algorithms, Sorting algorithms (selection, insertion, merge, quick, bubble, heap, shell, radix, bucket), queue, dequeuer, priority queues (linked and array implementations of queues), binary search for sorted arrays, hashing and indexing, open addressing and chaining, trees and tree traversals, binary search trees, heaps, graphs, breadth-first and depth-first traversal, shortest path, adjacency matrix and adjacency list implementations.			
Mapping of CLOs to PLOs			
PLOs/CLOs	CLO1	CLO2	CLO3
PLO1 (Academic Education)			
PLO2 (Knowledge for Solving Computing Problems)	✓		
PLO3 (Problem Analysis)		✓	✓
PLO4 (Design/Development of Solutions)		✓	
PLO5 (Modern Tool Usage)			
PLO6 (Individual and Teamwork)			
PLO7 (Communication)			
PLO8 (Computing Professionalism and Society)			
PLO9 (Ethics)			
PLO10 (Life-long Learning)			

Resources	
<u>Text Book:</u>	
Data Structures and Algorithms in C++ by Michael T. Goodrich, Roberto Tamassia, and David Mount, 2nd Edition, 2011, ISBN: ISBN: 13978-0470383278	
<u>Reference Books</u>	
- Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss (4th Edition), 2013, ISBN: 978-0132847377 - Malik, D S. Data Structures in C++ by Cengage Learning, 2nd Edition, 2009, ISBN: 978-0324782011	

Weekly Course Plan		
WEEK No.	Topics Covered	Lecture No.
WEEK 01	Introduction to the course	L1
	- Basic programming concepts - Introduction to Data Structures	L2
WEEK 02	Abstract Data Types and Applications	L3
	Analysis Tools - Complexity analysis - Asymptotic notations - Big Oh, Omega and theta notations	L4
WEEK 03	Linear Arrays - Introduction to Linear Arrays - Memory addressing scheme in Linear Arrays - Algorithm for Insertion, searching, traversing and deletion in linear arrays	L5
	Multidimensional Arrays - Multidimensional Array (2D array) - Memory addressing scheme in 2-D Array - Algorithm for addition and multiplication in 2-D array	L6
WEEK 04	List Reviving Structs and pointers	L7
	- Introduction to Singly linked list - Memory addressing scheme of Singly linked list - Algorithm for insertion, deletion, searching and traversing in singly linked list using pointers	L8

WEEK 05	Algorithm for insertion, deletion, searching and traversing in singly linked list using pointers	L9
	List - Introduction to Doubly linked list - Memory addressing scheme of Doubly linked list - Algorithm for insertion, deletion, searching and traversing in doubly linked list using pointers	L10
WEEK 06	Circular linked list	L11
	Stacks - Introduction to Stacks - Implementation of stack using arrays and linked list - Applications of stacks	L12
WEEK 07	Infix to prefix and postfix conversion	L13
	- Reversal of strings - Checking validity of brackets - Infix to postfix using stacks - Evaluation of postfix expression using stack	L14
WEEK 08	Queues - Introduction to Queue data structure - Implementation of Queue using array and linked list - Insertion and deletion in linear queue	L15
	- Insertion and deletion in circular queue - Priority queue and its types - Double ended queues	L16
WEEK 9	MID SEMESTER EXAM	
WEEK 10	Trees - Introduction to tree data structure - Basic terminologies related to tree - Representation of tree using array and linked list	L17
	- Types of tree - Tree traversal techniques (Pre-order, In-order, Post-order)	L18
WEEK 11	Trees - Formulation of binary tree from its traversals - Formulation of tree using infix expression	L19

	• Introduction to Binary search tree (BST) • Insertion, deletion and searching in BST	L20
WEEK 12	Heaps • Foundations, Binary Heaps • Binary Heaps, Heap Sort	L21
	• Insertion Sort • Selection Sort	L22
WEEK 13	Sorting • Merge Sort	L23
	• Quick Sort	L24
WEEK 14	Hashing • Introduction to hashing • Hash Functions	L25
	• Chaining, Open Addressing	L26
WEEK 15	Graphs • Basics Data Structures for Graphs	L27
	Graph Traversals • Depth First Search • Breadth First Search	L28
WEEK 16	Shortest Paths Algorithm • Dijkstra's Algorithm	L29
	Minimum Spanning Trees • Kruskal's Algorithm • The Prim's Algorithm	L30
WEEK 17	Project Presentations	L31
		L32
	END SEMESTER EXAMS	

DATA STRUCTURES - LAB	
Course Code:	CS-201
Credit Hours:	4(3+1)

Pre-requisite: CS-112			
Lab Objectives			
The key objectives of this course include:			
- To understand the design of fundamental data structures as well as algorithms that operate on them. - To understand the fundamental tradeoffs in the design of the data structures - To introduce tools for analyzing the time and space complexity of data structures			
Course Learning Outcomes			
After successful completion of this course, the students should be able to:	PLO	BT Level	
1. Implement the fundamentals of data structure as well as algorithms that operate on them	1,2	C2	
2. Apply and design various data structures methods for specified applications.	3, 4	C3	
3. Manipulate the fundamentals of data structure to solve domain specific problem.	3	C4	
Lab Contents			
Abstract data types, complexity analysis, Big Oh notation, Arrays and its operations, linked list & its various types, Stacks (linked lists and array implementations), divide and conquer algorithms, Sorting algorithms (selection, insertion, merge, quick, bubble, heap, shell, radix, bucket), queue, dequeuer, priority queues (linked and array implementations of queues), binary search for sorted arrays, hashing and indexing, open addressing and chaining, trees and tree traversals, binary search trees, heaps, graphs, breadth-first and depth-first traversal, shortest path, adjacency matrix and adjacency list implementations.			
Mapping of CLOs to PLOs			
PLOs/CLOs	CLO1	CLO2	CLO3
PLO1 (Academic Education)			
PLO2 (Knowledge for Solving Computing Problems)			
PLO3 (Problem Analysis)		✓	✓
PLO4 (Design/Development of Solutions)	✓	✓	
PLO5 (Modern Tool Usage)			
PLO6 (Individual and Teamwork)			
PLO7 (Communication)			
PLO8 (Computing Professionalism and Society)			
PLO9 (Ethics)			
PLO10 (Life-long Learning)			
Resources			
<u>Text Book:</u>			
Data Structures and Algorithms in C++ by Michael T. Goodrich, Roberto Tamassia, and David Mount, 2nd Edition, 2011, ISBN: ISBN: 13978-0470383278			

Reference Books

- Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss (4th Edition), 2013, ISBN: 978-0132847377
- Malik, D S. Data Structures in C++ by Cengage Learning, 2nd Edition, , 2009, ISBN: 978-0324782011

Weekly Lab Outline

WEEK No.	Topics Covered	<u>Lab No.</u>
WEEK 01	Basic programs in C++, Variable Declarations and Arithmetic Operations	L1
WEEK 02	Implementation of Selection Statement Using Dev C++, Implementation of Repetition Statement Using Dev C++	L2
WEEK 03	Introduction to Arrays, Insertion, Searching, Linear Search, Deletion and Sizeof()	L3
WEEK 04	Multidimensional Arrays, Multiplication of Matrix, Address-of operators, Pointer variables, Dynamic Memory, Allocation, Pointer to void, Pointers and Arrays, Structures and arrays, Structure within structure, Searching and Sorting in structure	L4
WEEK 05	Linked list Algorithms Implementation	L5
WEEK 06	Doubly Linked list Algorithms Implementation	L6
WEEK 07	Stacks implementation via arrays and linked list and various operations of stacks	L7
WEEK 08	Lab Viva + (Sessional)	
WEEK 9	Queues, Array based Implementation, Linked list based Implementation, Circular queue based implementation	L8
WEEK 10	Priority Queues Implementation and Trees creation	L9
WEEK 11	Binary Search Tree Creation and finding min and max items	L10
WEEK 12	Vectors, Heaps array-based implementation, max and min heap	L11
WEEK 13	Graphs representation, Adjacency Matrix, Adjacency list, and Breath first traversal.	L12
WEEK 14	Depth First Search	L13
WEEK 15	Sorting Algorithms : Quick Sort, Shell Sort, Radix Sort ,Insertion Sort	L14
WEEK 16	Final Exam Lab Viva Voce	